



2026 Chantilly Math Competition

Middle School Division - Team Round

Competition Instructions

1. Do not open the test until instructed.
 2. No computational aids other than pencil/pen are permitted.
 3. Write answers in designated boxes on answer sheet.
 4. All answers must be nonnegative integers: $\{0, 1, 2, \dots\}$.
 5. Points per question range from 9 to 16.
 6. The test consists of 16 short-answer questions to be solved in 60 minutes.
 7. The final question will be used for tie-breaking.
 8. There is no penalty for incorrect answers.
 9. You will be graded on your answers—not your work.
 10. If you have any questions, ask your proctor.
 11. **Notation/Definitions:**
 - The dot $\cdot$ denotes multiplication: $a \cdot b = a \times b$.
 - $n!$ denotes the factorial of n : $n! = n \times (n - 1) \times \dots \times 1$.
 - Two integers a and b are *relatively prime* if and only if $\gcd(a, b) = 1$.
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1. (9 pts) How many subsets of $\{1, 2, 3, 4, 5, 6, 7\}$ have sum greater than 14?
2. (9 pts) Given that

$$((x^2 - 5x + 9)^2 - 5(x^2 - 5x + 9) + 9)^2 - 5((x^2 - 5x + 9)^2 - 5(x^2 - 5x + 9) + 9) + 9 = x$$

for some real number x , compute x .

3. (10 pts) A palindrome is a positive integer that has the same value when its digits are reversed. Compute the smallest positive integer n such that n^2 is a palindrome with an even number of digits.
4. (10 pts) Shubham is at some uniformly random point (a, b) where a and b are both integers between -2 and 2 inclusive. Every minute, Shubham moves to a point with integer coordinates that is exactly $\sqrt{5}$ away from the point he is currently at. What is 100 times the expected value of the smallest possible number of minutes until Shubham is at $(0, 0)$?
5. (11 pts) Define $\tau(n)$ as the number of positive integer divisors of n . Compute the sum of all integers n such that $1 \leq n \leq 100$ and $2\tau(\tau(n)) = \tau(n)$.
6. (11 pts) There are 6 cities, every pair of them is connected by a road, and it is impossible to switch roads without arriving at a city. It is snowing, so every hour, 1 of these roads is closed. After 2 hours, the probability that it is possible to start in some city and drive along all the non-closed roads, each exactly once, can be expressed as $\frac{a}{b}$ for relatively prime positive integers a and b . Compute $10a + b$.
7. (12 pts) Compute the largest of the 5 distinct primes that are divisors of $2^{41} + 2^{67}$.
8. (12 pts) Let $ABCD$ be a parallelogram such that $AB = 41$ and $BC = 82$. Let O be the intersection of AC and BD . Let T and I lie on AD and AC , respectively, so that they both lie on the internal angle bisector of $\angle ABC$. Given that $AC = 67$, compute the area of parallelogram $ABCD$ divided by the area of quadrilateral $DOIT$.
9. (13 pts) How many nonnegative integers have at most 6 digits, have no digit larger than 7, and contain the digit 6 immediately followed by the digit 7 somewhere inside it?
10. (13 pts) All 4 faces of a tetrahedron have side lengths of 6, 7, and 7. Given that the volume of this tetrahedron can be expressed as $a\sqrt{b}$ where a and b are positive integers and b isn't divisible by the square of a prime, compute $a + 10b$.
11. (14 pts) Soham thinks of some positive integer n . How many distinct nonnegative integers could be the remainder when n^n is divided by 2025?
12. (14 pts) Compute the maximum possible value of

$$2025 \cdot \frac{x^4 \cdot 4^x}{2^{x \cdot 6} + 2x^6}$$

over all real numbers x .

13. (15 pts) Jonathan has a deck of 52 cards. Each card has 1 number on it. For each of the numbers 1, 2, 3, and 4, there are 13 cards in the deck with that number on it. A group of 4 cards is called awesome if and only if no 2 cards in the group have the same number. Jonathan splits his deck of 52 cards into 13 disjoint groups of 4 cards. Given that the expected number of these 13 groups that are awesome can be expressed as $\frac{m}{n}$ for relatively prime positive integers m and n , compute $m + n$.
14. (15 pts) Ray has an integer $t = 0$, and is facing north. Every minute, Ray turns t degrees clockwise, and if he is facing north after that turn, he increases t by 1. In how many minutes will $t = 360$?
15. (16 pts) Jake draws a hexagon. All its angles are equal. All its side lengths are either 6 or 7. The sum of all distinct possible areas of this hexagon is $\frac{m}{n}\sqrt{3}$ where m and n are relatively prime positive integers. Compute $m + n$.
16. (16 pts) For all non-negative integers d , let P_d be the unique polynomial of degree d such that

$$P_d(2^k) = 2^{k+1}$$

for all non-negative integers k less than d , and $P_d(2^d) = 1$.

Given that

$$P_0(5) + P_1(4) + P_2(3) + P_3(2) + P_4(1) + P_5(0)$$

can be expressed as $\frac{a}{b}$ for relatively prime positive integers a and b , compute $a + b$.

- TB. (Tiebreak) Ryan has a random number generator with 2 integer parameters $1 = m < M = 67$. While $M - m > 1$, the random number generator picks some integer between m and M exclusive, and Ryan changes either m or M , each option with probability $\frac{1}{2}$, to that integer. Estimate the integer nearest to 1 billion times the probability that $m = 41$ when $M - m = 1$.

2026 Chantilly Math Competition - Answer Sheet
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Team Name: _____

Team Member 1: _____

Team Member 2: _____

Team Member 3: _____

Team Member 4: _____

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

TB